MILITARY SPECIFICATION

POLYVINYL ACETATE AQUEOUS EMULSION (PAAE) (FOR USE IN AMMUNITION)

1. SCOPE

1.1 This specification covers one type of polyvinyl acetate aqueous emulsion and an aqueous solution of chromic nitrate catalyst for use as a binder in non-metallic cartridge cases.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-109 - Inspection Terms and Definitions

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR - 171-190 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, Government Printing Officer, Washington, D.C. Orders for the above publications should cite: "49 CFR 171-190 (latest revision)".

FSC: 1376

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3. REQUIREMENTS

3.1 Material.-The solids content of the polyvinyl acetate aqueous emulsion copolymer consists of 90 percent minimum by weight vinyl acetate, 2 to 5 percent N-methylol acrylamide and emulsified by 2 to 8 percent polyvinyl alcohol.

3.1.1 Total solids.-The total solids content of the emulsion shall be 50 to 53 percent by weight when determined as specified in 4.3.1.

3.1.2 Viscosity.-The viscosity of the emulsion shall be 4,500 to 7,500 centipoise (cps) when determined as specified in 4.3.2.

3.1.3 Baked insolubles.-The baked insoluble content of the emulsion shall be 94 percent minimum when determined as specified in 4.3.3.

3.2 Catalyst.-The catalyst shall be an aqueous solution of 43 to 46 percent chromic nitrate.

3.2.1 Color.-The color of the aqueous solution shall be a dark green.

3.2.2 Specific Gravity.-The specific gravity of the aqueous solution of the chromic nitrate shall be 48.5 to 49.5 degrees baume when determined as specified in 4.3.4.

3.3 Workmanship.-The polyvinyl acetate aqueous emulsion and the chromic nitrate aqueous solution shall be free from impurities such as wood chips, oil, and solvents not specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

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4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Drawing, specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot", as used throughout this specification, refers to an inspection lot, which is defined as en essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall respresent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot.

4.2.2 Testing

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4.2.2.1 Sample.-A random sample of 10 containers shall be selected from each lot, all containers being of the same size. When lots are comprised of 10 containers or less, each container shall be sampled.

4.2.2.2 Preparation of composite samples.-Four-ounce samples of PAAE shall be removed from each of the ten containers selected to form a composite sample. The four ounces selected shall be blended to form two 20-ounce composite samples. The two samples shall be tested in accordance with the requirements of 3.1 through 3.3. If either of the composite samples fails to meet the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Total solid - Run in duplicate - Major defect, Code No. 01001 (see 6.2).-Weigh rapidly into a tared aluminum dish 1 to 2 grams of the emulsion. Place the dish and sample in a 130°C oven for 30 minutes. Remove and cool the dish and sample to room temperature in a dessicator. Reweigh the dish and sample. Place in the oven again and repeat above until constant weight is obtained.

$$\frac{\text{Dried Solids Weight}}{\text{Initial Sample Weight}} \times 100 = \text{Percent Solids}$$

4.3.2 Viscosity.-Run in Duplicate, Major defect, Code No. 02001.-The viscosity shall be determined in accordance with ASTM Method D 1084, Method B, using a No. 4 spindle at 20 RPM at 72°F.

4.3.3 Baked insoluble (Run in duplicate) - Major defect, Code No. 03001.-A six mil film shall be cast on a glass plate. After the film is dried, the film shall be stripped and placed in between release paper. The film shall then be cured at 25°C and a relative humidity of 50 ± 2 percent. A 0.5 gram sample of the cured film shall be weighed in a 250 ml Erlenmeyer flask, and 100 ml of acetone added. The solution shall be refluxed for 2 hours. The solution shall be cooled and a 20 ml portion shall be pipetted into a tared weighing dish. The solvent shall be evaporated using a steam bath. The dish then shall be placed in a 130°C draft oven for 30 minutes or until a constant weight is obtained. Calculate the percent insoluble as follows:

$$\text{Percent insoluble} = 100 \frac{(1 - \text{residue weight} \times 5)}{\text{Initial film weight (gms)}}$$

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4.3.4 Specific gravity (run in duplicate) - Major defect, Code No. 04001.-The specific gravity shall be determined by a standard Baume hydrometer for the range specified in paragraph 3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Packing.-Polyvinyl acetate aqueous emulsion shall be packed and shipped in such a manner as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Containers shall conform to the ICC Regulation.

5.2 Marking.-Marking to insure safe handling shall conform to the ICC Regulations stipulated in the Code of Federal Regulations.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Test results.-A copy of the test results should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, New Jersey 07801.

6.4 Polyvinyl acetate aqueous emulsion.-An approved source of supply for PAAE is National Starch and Chemical Corporation, 750 Third Avenue, N.Y., New York, 10017.

Custodian:
Army-MU

Preparing Activity:
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